

Recognition of Shock

Now we are going to bring our discussion of shock to the next level and start to explore some patterns that can help us with recognition of shock. Remember, if we no longer define shock as low blood pressure, we need to have some indicators to help us recognize what can be a startlingly subtle yet dangerous state.

But first, a caveat. Imagine you go to the beach one warm summer day. Even though it is a popular beach, you note that there are almost no people there. You notice a purple flag flying over the lifeguard stand. You then notice people running out of the water. Maybe this pattern means that there is something dangerous in the water. Or maybe it is just early in the day, most people haven't arrived yet, and the people running out of the water were just playing.

The point being that patterns can be helpful, but they can also be limiting. They don't always hold true, but they can give a framework for what to expect, especially if you have an appreciation for the limitations and exceptions.

That said, let's consider a pattern commonly seen in shock.

WHAT DOES SHOCK LOOK LIKE AT THE BEDSIDE?

Imagine you are receiving a patient in the intensive care unit (ICU). The team that is handing him off says, "I'm not sure what is going on with him, but he has us worried. We are thinking he may be in shock."

What might you expect this patient to look like over the next 12–24 hours assuming he continues to worsen? Not every presentation is the same, but the following is a pattern that is useful to consider (Fig. 3.1).

Let's consider each of these phases.

Phase 1: cardiac output compensation

Remember that as $\text{DO}_2:\text{VO}_2$ falls, the most ready way that the body can increase this is by augmenting cardiac output. Cardiac output increases when the stroke volume increases (heart squeezes harder) or heart rate increases (heart beats faster). In most people, this can augment CO, and consequently DO_2 several fold (up to 35-fold in some athletes).

Clinically, this can be manifested by tachycardia or by increased pulse pressure (the difference of systole and diastole) (Fig. 3.2).

Ultimately, if DO_2 is not able to keep up with VO_2 requirements, then the $\text{DO}_2:\text{VO}_2$ ratio starts to fall, manifested by lower venous oxygen saturation (Fig. 3.3).

Phase 1: Cardiac output compensation

Phase 2: Decreased renal function

Phase 2: Decreased renal function

Phase 3: Lactic acidosis

Phase 4: Liver injury

Clinical presentation of shock

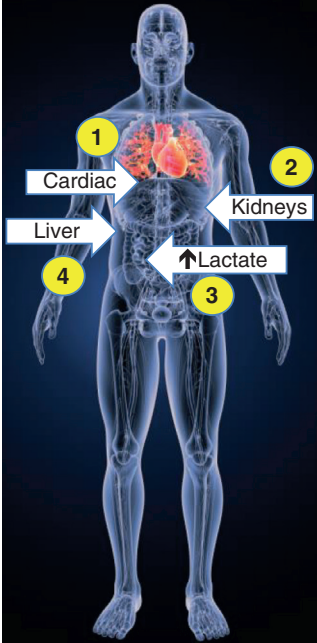


FIG. 3.1 A commonly observed pattern in shock: augmentation of cardiac output, diminished renal function, lactic acidosis, and liver injury. (Modified from [Sciencepics/Shutterstock.com](https://www.shutterstock.com).)

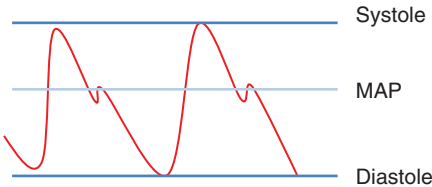


FIG. 3.2 Pulse pressure: the difference between systole and diastole. MAP, Mean arterial pressure.

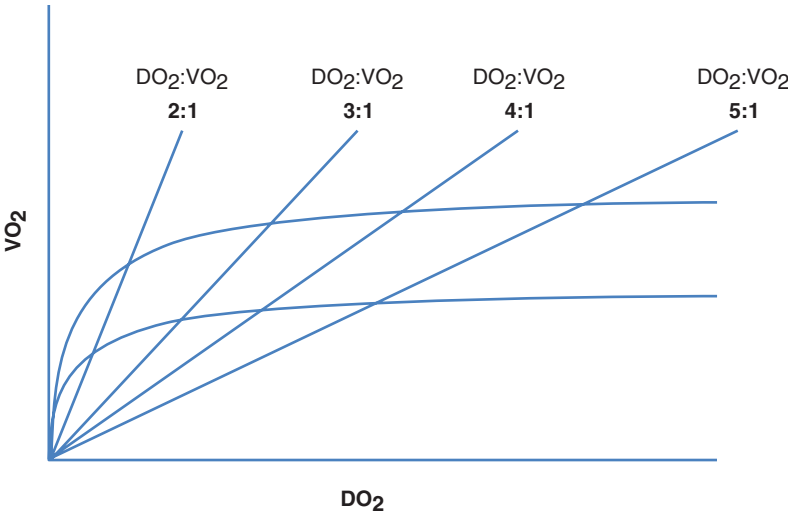


FIG. 3.3 Venous oxygen saturation and DO₂:VO₂.

When does this pattern not apply?

There are many reasons other than shock for the heart rate/cardiac output to be augmented in patients in the ICU (agitation, delirium, medication effect, withdrawal, etc.). However, a rise in these parameters should give pause and lead to the consideration that it may be compensation for a drop in DO_2 .

Additionally, a venous oxygen saturation cannot always be used as a surrogate for DO_2 : VO_2 . We remember that if the VO_2 remains constant, a dropping venous oxygen saturation can be a surrogate for a drop in DO_2 : VO_2 .

However, what happens when the cardiac output remains in a high output state, say due to a catecholamine surge or in early sepsis? In this case, the blood may be circulated around the periphery so fast that the oxygen is unable to be extracted. This may lead to a higher than expected venous oxygen saturation, which may not correspond with adequate DO_2 : VO_2 .

In another example, as shock progresses, target organs may begin shutting down, leading to a decreased VO_2 . This will lead to an elevated DO_2 : VO_2 ratio that does not actually mean that the body is adequately delivering oxygen to meet the tissue needs, but rather may have a more ominous meaning (Fig. 3.4).

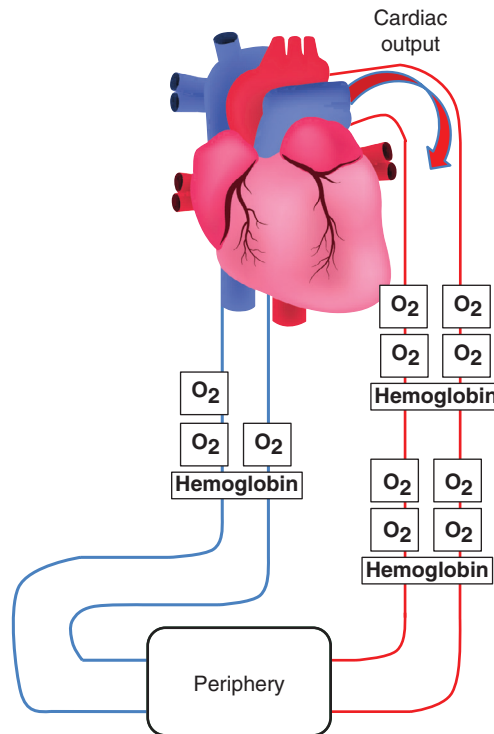


FIG. 3.4 High venous oxygen saturation as a consequence of decreased VO_2 in decompensated shock. (Modified from [Ody_Stocker/Shutterstock.com](#).)

Phase 2: decreased renal function

As DO_2 : VO_2 continues to fall, the renal function will often decline next, manifested by a drop in urine output and an eventual rise in creatinine (Fig. 3.5). This is due to a variety of mechanisms, some adaptive, some pathologic.

The adaptive component of this mechanism is a drop in urine output in response to a decrease in renal perfusion pressure. This has the additional effect of maintaining intravascular volume and augmenting preload. However, there can also be a pathologic mechanism, in which decreased

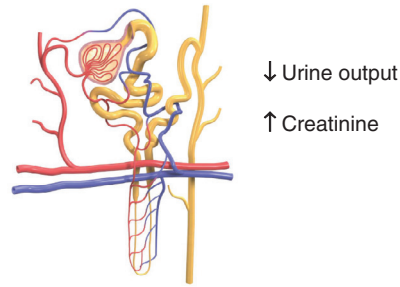


FIG. 3.5 Renal effects of decreased perfusion. (Modified from Axel_Kock/Shutterstock.com.)

perfusion and shock state can cause damage to the renal glomerular architecture. This can be manifested as a worsening urine output and rising creatinine that does not improve after perfusion is restored.

Phase 3: lactic acidosis

Lactate level is a common test ordered to quantify patients in shock. The reason is that lactate is a marker of a transition from aerobic metabolism to anaerobic metabolism. Therefore, as $\text{DO}_2:\text{VO}_2$ continues to drop, the proportion of cells requiring anaerobic metabolism increases and lactic acid rises. This rise can be delayed and typically happens after the compensation in cardiac output and the decline of renal function.

Does an elevated lactate always mean a patient is in shock?

To better understand how to interpret a lactic acid level in shock, let's go all the way down to the cellular level and consider how glucose is metabolized.

Glucose is transported into the cell via the GLUT4 receptor and enters the cytoplasm. It is then phosphorylated and catalyzed by the enzyme G6PD. The phosphorylated glucose is then split into pyruvate generating adenosine triphosphate (ATP). Pyruvate is then transported into the mitochondria via the enzyme pyruvate dehydrogenase.

Once in the mitochondria, the pyruvate is used to generate ATP via the electron transport chain with oxygen playing a central role by being the final electron acceptor (Fig. 3.6).

Stay with me! There is a reason for every receptor and enzyme that was just mentioned.

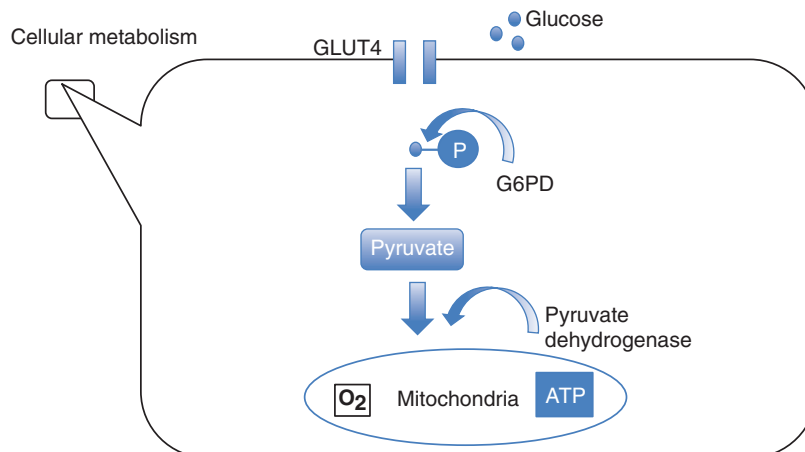


FIG. 3.6 Normal metabolism of glucose. *ATP*, Adenosine triphosphate.

What happens if there is no oxygen? (Decreased $\text{DO}_2\text{:VO}_2$)

If there is a decreased $\text{DO}_2\text{:VO}_2$, then pyruvate accumulates and it is metabolized into lactate (Fig. 3.7). This forms the basis for the assumption that lactic acidosis is an indicator of shock. Since it takes time for reserves to be depleted and for cells to shift to this anaerobic metabolism, this lactic acidosis usually occurs on the order of hours, rather than the seconds as seen for cardiac compensation and the minutes seen for renal compensation.

Ok, so the higher the lactate, the worse the $\text{DO}_2\text{:VO}_2$, right?

This is the theory to some degree, but again, we need to understand when it works and when it breaks down. Let's explore further.

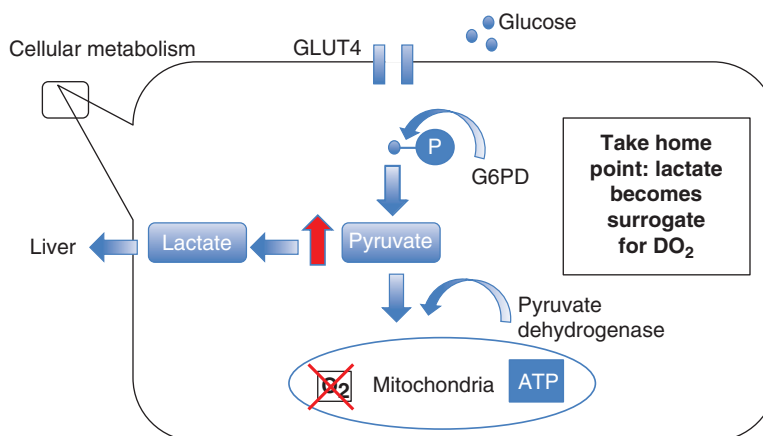


FIG. 3.7 Impaired oxygen delivery and the production of lactate. *ATP*, Adenosine triphosphate.

When elevated lactate does not necessarily mean a deficiency in $\text{DO}_2\text{:VO}_2$

Let's examine when lactate level may not necessarily correspond to a deficiency in DO_2 (or at the very least, when augmenting DO_2 may not necessarily lead to a **clearance** of lactate). As an example, let's consider septic shock.

In sepsis, there may be a variety of derangements to include hypoglycemia, downregulation of GLUT4 through catecholamine administration, decreased availability of phosphorous, and decreased activity of pyruvate dehydrogenase. All of these derangements may contribute to altered levels of pyruvate in the cytoplasm of cells of septic patients, independent of oxygen delivery/availability. Coupled with liver injury, which can also predominate in sepsis, you can imagine how elevated lactate/impaired lactate clearance can occur, even in the setting of adequate oxygen delivery (Fig. 3.8).

Phase 4: liver injury

Liver injury is the fourth phase of shock that can be very revealing. While augmentation of cardiac output (phase 1) can occur in a matter of seconds, decreased renal function can occur in a matter of minutes, and lactic acidosis can occur over the course of hours, liver injury usually becomes apparent over the next 1–2 days.

The pattern is usually a precipitous rise in aspartate aminotransferase/alanine aminotransferase (AST/ALT) over the course of a few days. In severe shock, these values may elevate to the thousands in the first day or so, with improvement after the shock is addressed. In the same way that elevation of liver function tests can be delayed in shock liver, the improvement in these parameters can also be delayed after the cause of shock is resolved.

The process can be subtle and difficult to pick up. For example, there are other causes of elevated liver function tests in patients with shock, such as sepsis itself, right heart dysfunction with

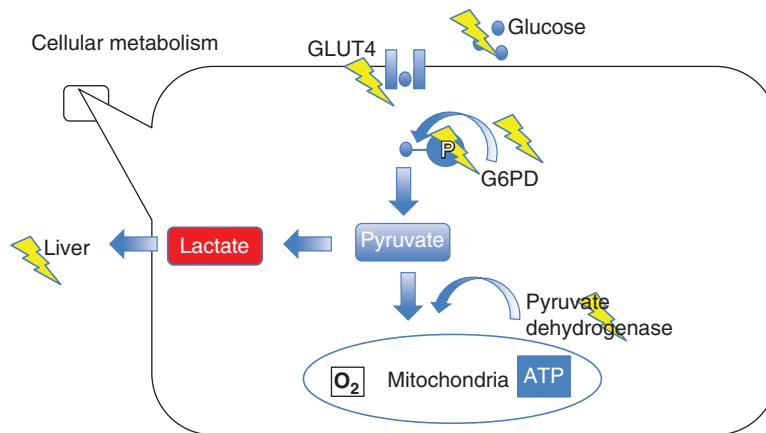


FIG. 3.8 Impaired glucose metabolism and increased lactate production in sepsis. *ATP*, Adenosine triphosphate.

congestion of blood in liver, acalculous cholecystitis, and medication effects. Shock liver can often be distinguished from these other causes of liver injury by the degree of elevation of liver function tests, but this is not always the case. Liver injury in the setting of shock should always be put in the appropriate clinical context.

PUTTING IT ALL TOGETHER

Shock can be overt in its presentation or very subtle. Regardless, time is often of the essence – even small delays in the recognition of shock can lead to a downward spiral that can be difficult to reverse. In the next chapter, we explore the importance of assessing the cause of decompensation and quantifying the response of the patient to our interventions. Early and accurate recognition will be an essential part of this process.

SUGGESTED READING

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